PRODUCT OVERVIEW & PROPOSAL

ADEM: Smart Exam Evaluation Platform

How the platform works for schools, teachers and parents: workflows, features, system design, pricing model and the decisions we need from you.

PRODUCT

ADEM: AI-assisted Digital Examination Management & Evaluation

DOCUMENT

Product Overview & Proposal v1.0

BOARD / CLASSES

CBSE · Class 10 and Class 12

DATE

30 October 2026

Prepared for review and sign-off. Please answer the questions in section 11.

Contents

01 Executive summary

02 Who uses ADEM

03 Use case diagram

04 Workflow diagrams

05 System architecture

06 Features for each user

07 How credits work

08 Safety, privacy and quality

09 Delivery plan

10 Glossary

11 Questions, blockers and issues

01 Executive summary

ADEM marks handwritten exam answer sheets using artificial intelligence. Teachers today use it to upload a question paper, an answer key and students' answer sheets, and ADEM suggests marks for every question, which the teacher approves.

We are now growing ADEM from a teachers' tool into a complete platform with four kinds of users:

RUNS THE PLATFORM
Platform Admin

Looks after every school and parent account, sets prices, handles payments and support.

OVERSEES A SCHOOL
School Management

Signs the school up, invites teachers, and can see every exam, mark and report in the school.

RUNS EXAMS
Teacher

Creates exams, uploads answer sheets and approves the AI's marks. Works the same way as today.

PRACTISES AT HOME
Parent

Signs up independently and sets exams for their children. ADEM creates the paper, times the exam and marks the answer sheet.

What changes for you

- **Schools join by themselves** through a sign-up page. They go live once the Platform Admin approves them.
- **Parents become customers too.** Children can sit timed CBSE practice exams at home and get marked results.
- **Question papers are created automatically** from CBSE textbooks and the CBSE paper pattern, for any chosen lesson or unit.
- **Simple pay-as-you-go pricing using credits,** instead of fixed subscriptions.
- **Sign in with Google** for every user, as well as email and password.
- **Better Tamil marking accuracy,** delivered as part of this work.

02 Who uses ADEM

User	How they join	What they can do
Platform Admin	Created by the platform team	Manage all schools and parents, approve new schools, set credit prices, see payments, resolve complaints about marks, answer support requests, see a full activity log.
School Management	Signs up on the website; one login per school	Invite and manage teachers, see every exam, mark, answer sheet and report card in the school (view only), request question papers, buy credits.
Teacher	Invited by email by their school	Everything teachers do today: classes, students, exams, uploading papers and answer sheets, approving AI marks, results and report cards.
Parent	Signs up on the website; not linked to any school	Add several children, choose a subject and unit, let the child sit a timed exam, upload the answer sheet, see the score and the marked sheet, buy credits.

> **One email, one role**
> Each email address belongs to one type of user. A teacher who is also a parent uses a different email for the parent account.

03 Use case diagram

The diagram shows each type of user and the main things they do on the platform.

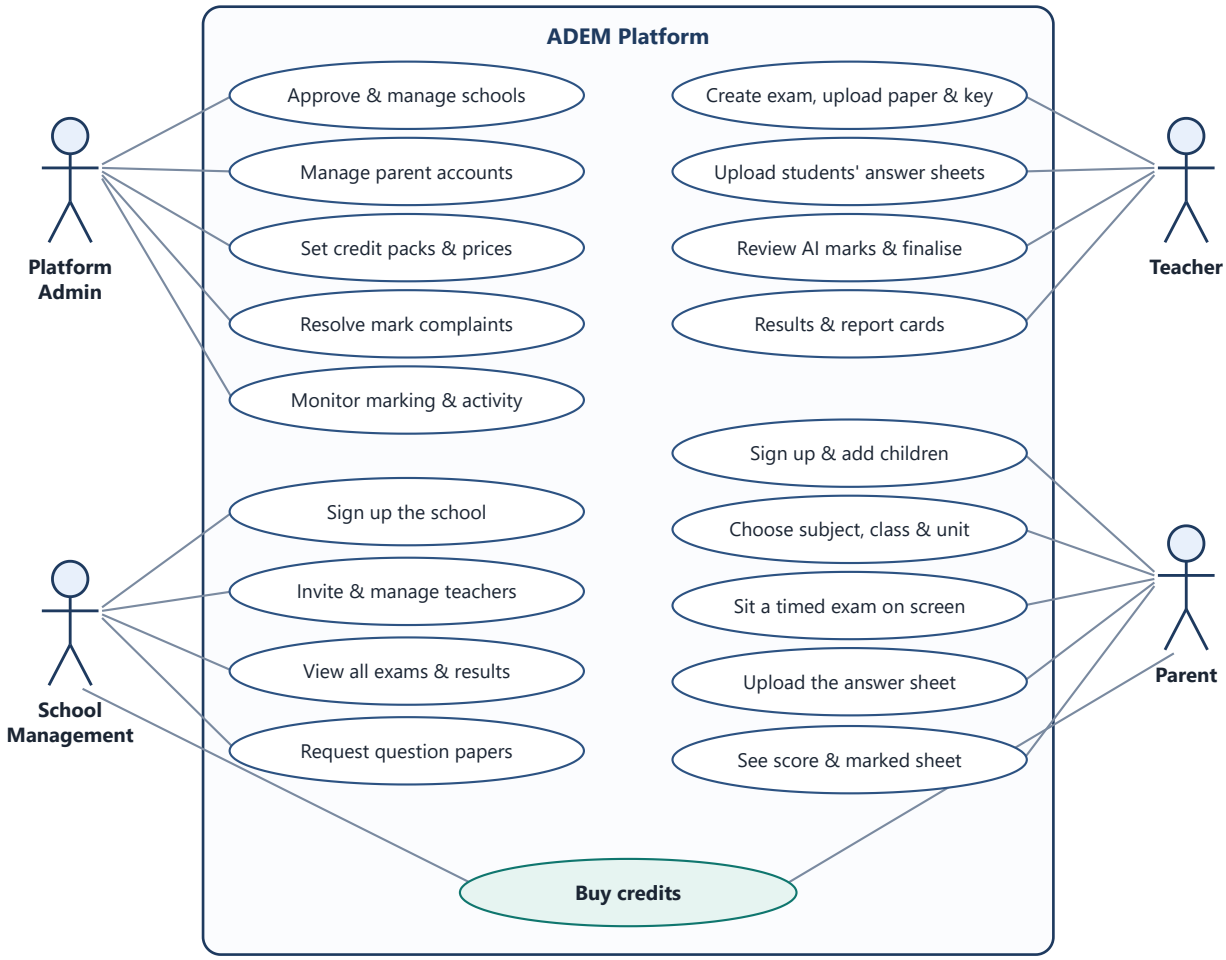


Figure 1. Use cases. All users sign in with email and password, or with Google.

04 Workflow diagrams

Done by a person Done automatically by ADEM Done by the Platform Admin

4.1 School and teacher workflow

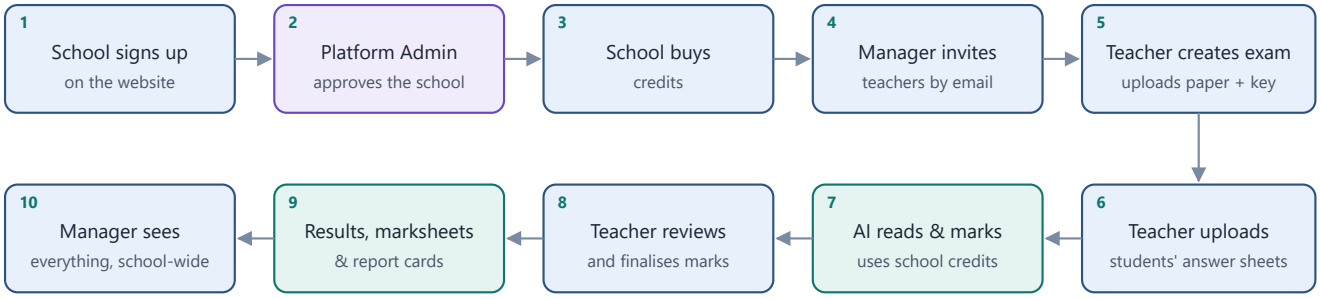


Figure 2. From school sign-up to report cards.

4.2 Parent exam workflow

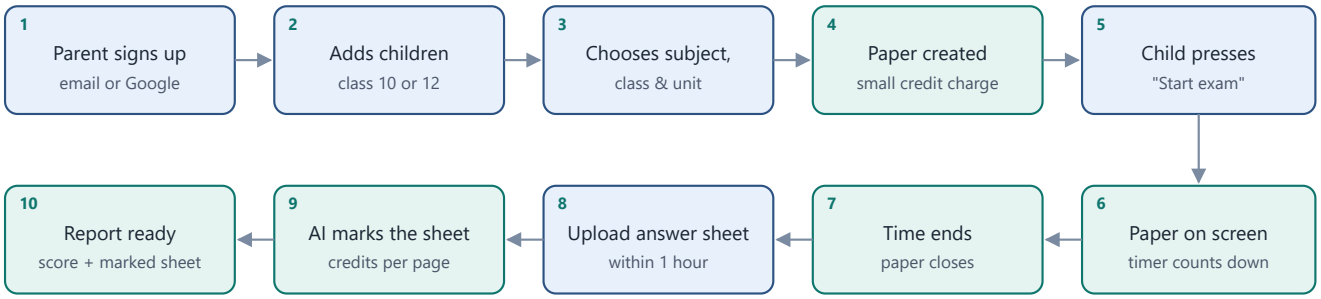


Figure 3. A parent-set exam, from choosing a unit to the marked report.

- The paper is shown **on screen only**. It cannot be downloaded or printed, and it closes when time is up.
- The timer keeps running even if the browser is closed. There is no pause.
- The **answer key stays hidden**. The report shows the score and the marked answer sheet.
- If no answer sheet is uploaded within 1 hour, the exam is marked as missed and no marking credits are used.
- The child can retake the unit with a fresh paper.

4.3 How question papers are created

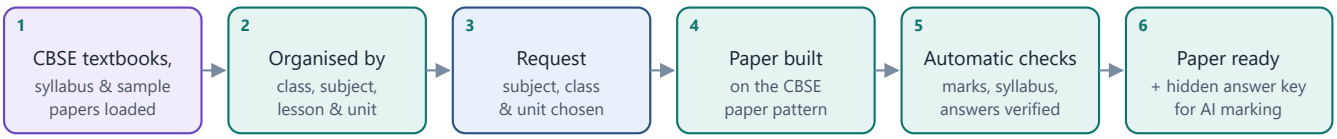


Figure 4. Automatic question papers, grounded in CBSE textbooks.

Every question is built from real textbook content, so the paper stays within the CBSE syllabus. Before parents can use the generator, a subject expert reviews a sample of the papers it produces.

05 System architecture

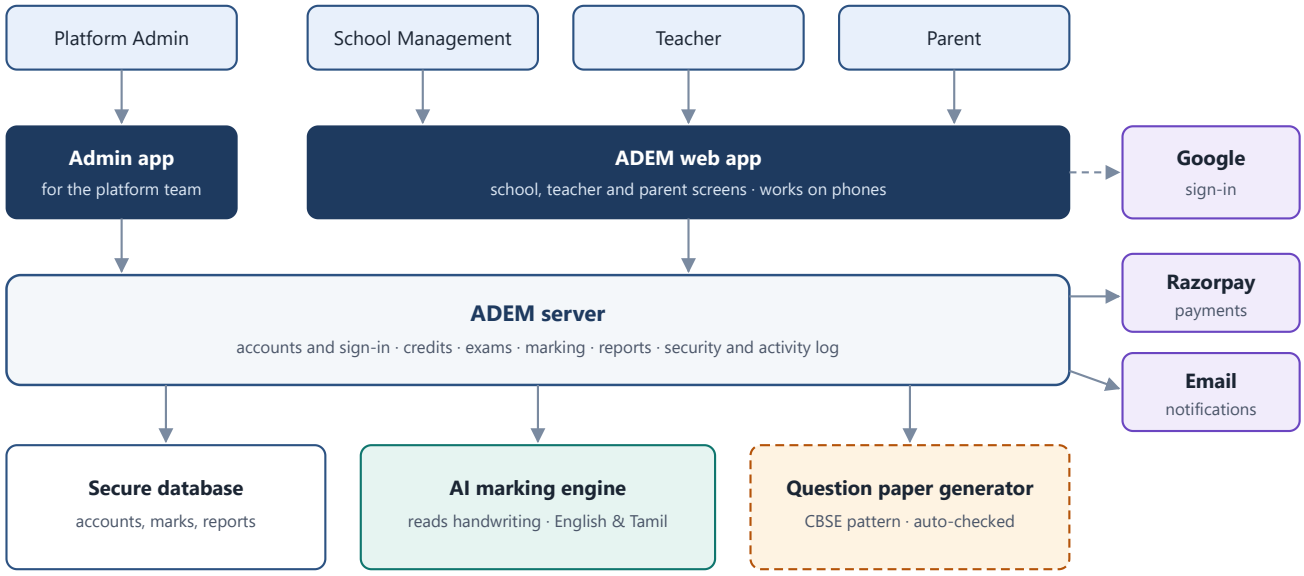


Figure 5. How the parts of ADEM fit together. Amber dashed = new component.

Part	What it does, in plain terms
Admin app	A separate, private website for the platform team to run the business.
ADEM web app	The website schools, teachers and parents use. It adapts to phones, so parents can photograph and upload answer sheets easily.
ADEM server	The central system that checks who is signed in, keeps each school's and parent's data separate, tracks credits, and runs exams and reports.
Secure database	Stores accounts, exams, marks and reports.
AI marking engine	Reads handwritten answers (including Tamil) and marks them against the answer key.
Question paper generator	Creates CBSE-pattern papers from textbook content and checks them automatically.
Google, Razorpay, Email	Trusted outside services for "Sign in with Google", online payments and emails.

06 Features for each user

PLATFORM ADMIN

Run the platform

- Dashboard: schools, parents, marking activity, credits and revenue
- Approve, suspend or remove schools and parents
- Help users: reset passwords, unlock accounts, sign in as them for support
- Set credit packs, prices and costs; add or remove credits by hand
- Payments, refunds and GST invoices
- Watch AI marking and retry anything that failed
- Review complaints about marks and refund credits
- Answer support requests
- Upload textbooks and sample papers for the paper generator
- Full activity log and admin staff accounts

SCHOOL MANAGEMENT

Oversee the school

- Sign up online, with email or Google
- Dashboard: exams, progress, averages, pass rates, toppers
- Invite, deactivate and help teachers
- See every exam, mark, answer sheet and report card (view only)
- School-wide reports and comparisons
- Request question papers
- School profile, logo and report card design
- Year-end rollover of classes
- Buy credits, see usage, download invoices

TEACHER

Run exams

- Join by email invitation; sign in with email or Google
- Classes, subjects and students, with import and export
- Create exams; upload the paper, answer key and answer sheets
- See the credit cost before marking starts
- AI marking; review and correct (re-marking is free)
- Results, marksheets and report cards

PARENT

Practise at home

- Sign up online, with email or Google
- Add several children (class 10 or 12)
- Choose subject, class and lesson or unit
- Timed exam on screen
- Upload the answer sheet by photo or PDF, with a photo quality check
- Score, marked answer sheet, strengths and weak chapters
- Progress over time; retakes
- Buy credits; raise a complaint about marks; delete my data

For everyone

- Sign in with email and password, or Google; password reset by email; account lock after repeated wrong passwords.
- Email notifications: account verification, invitations, receipts, marking complete, low credits, upload reminders.
- Works on phones as well as computers.

07 How credits work

ADEM uses **credits**, a simple pay-as-you-go system. Schools and parents buy credit packs online through Razorpay and spend them as they use the platform.

Action	Who pays	Credits used
Buying a credit pack	School or parent	Adds credits; a GST invoice is issued
Choosing a question paper	Parent	A small fixed amount
AI marking of an answer sheet	School (teacher exams) or parent	Per page marked
Re-marking after a teacher's correction	No one	Free
Marking that fails for technical reasons	No one	Credits returned automatically
Upheld complaint about marks	No one	Credits refunded

> **Example (illustrative values only)**
>
> A parent chooses a Class 10 Science unit paper (5 credits). The child writes 8 pages, which are marked at 2 credits per page (16 credits). Total: 21 credits. The actual values are for you to decide (see section 11).

- Every school and every parent has a **credit wallet** with a full history of what was bought and used.
- New schools and parents receive some **free credits** to try the platform (once per email, phone number and school).
- Before marking starts, the user sees the cost and confirms it.

08 Safety, privacy and quality

Area	How ADEM handles it
Data separation	Each school sees only its own data. Each parent sees only their own children. Schools never see parents' data.
Secure sign-in	Protected passwords, Google sign-in, account lock after repeated wrong attempts, sign out from all devices.
New schools checked	A school is only activated after the Platform Admin approves it.
Children's data	Parents give consent at sign-up, as required by India's Digital Personal Data Protection Act 2023. Parents can ask for their data to be deleted. Uploaded files are removed after an exam is finalised.
Activity log	Every important action is recorded: who, what and when.
Marking quality	The answer key is always the reference. Teachers approve marks for school exams. Parents can raise a complaint about any mark.
Question paper quality	Papers come from CBSE textbook content and pass automatic checks. A subject expert reviews sample papers before launch.
Photo quality	Blurry, dark or rotated photos are caught before any credits are used, and the parent is asked to retake them.
Exam fairness	Papers are on screen only, timed, and close automatically. This discourages misuse, though no home exam can be fully supervised.

09 Delivery plan

Work runs in two parallel tracks. Each phase ends with something you can see and try.

Phase	What gets built	What you will be able to see
1	Foundations and accounts	New sign-up and sign-in for all users, Google sign-in, email invitations and password reset.
2	Credits and payments	Credit wallets, buying packs through Razorpay, invoices.
3	Admin app	The platform team's dashboard: schools, parents, credits, payments, complaints, activity log.
4	School and teacher	School dashboard, teacher invitations, full school view; teachers working on the new accounts.
5	Parent portal	Children, timed exams, uploads, marked reports.
6	Testing and launch	Full testing, security review, phone layout, go-live.
AI track	Tamil accuracy and paper generator	Better Tamil marking; automatic papers for pilot subjects, switched on for parents after expert review.

Included now, and to follow

First release

- All sign-up and sign-in, including Google
- Admin app, School Management, Teacher, Parent
- Credits, payments and invoices
- Parent timed exams with marked reports
- Improved Tamil marking

Follow-up release

- Detailed school comparison reports and exports
- Advanced analytics
- Automatic papers for more subjects
- School-level report card designer

10 Glossary

Term	Meaning
Answer key	The correct answers and marking points the AI marks against.
AI marking	The computer reads handwritten answers and suggests marks for each question.
Credits	Prepaid units used to pay for question papers and marking.
Credit wallet	A school's or parent's credit balance and history.
Platform Admin	The team that runs the ADEM platform.
School Management	The school-level account that oversees all teachers and exams.
Question paper generator	The part of ADEM that creates CBSE-pattern papers automatically from textbook content.
Razorpay	The online payment service used to buy credits.
Google sign-in	Signing in with an existing Google account instead of a new password.
DPDP Act 2023	India's Digital Personal Data Protection Act, which governs personal data, including children's.

11 Questions, blockers and issues

Questions to clarify

Please answer these so we can finalise pricing and setup. Where we have a suggestion, it is shown in green. You can simply reply "agree" to accept it.

Pricing and credits

1. **What credit packs and prices should we offer?**
   Suggestion: three packs (100, 500 and 1,000 credits) with a discount on the larger packs.

2. **How many credits should choosing a question paper cost a parent?**
   Suggestion: 5 credits.

3. **How many credits should AI marking cost per page?**
   Suggestion: 2 credits per page.

4. **Should Tamil marking cost more than English? It uses an extra AI reading step.**
   Suggestion: 1.5 times the normal rate.

5. **How many free credits should new schools and new parents get?**
   Suggestion: 50 for schools (after approval), 20 for parents.

6. **Should credits expire?**
   Suggestion: 12 months after purchase.

7. **What happens when a user runs out of credits?**
   Suggestion: block new marking and show a "Buy credits" button.

Question papers

8. **Which subjects should the automatic paper generator start with?**
   Suggestion: one or two Class 10 subjects, e.g. Science and Maths, then expand.

9 **Who will provide the source books and sample papers for those subjects?**
We already have some CBSE sample papers and marking schemes.

10 **Who will be the subject expert reviewing sample papers before launch?**

11 **If the paper generator is not ready at launch, should the parent exam feature show "coming soon"?**
Suggestion: yes. Everything else launches on time.

Setup and accounts

12 **Please share the Google sign-in Client ID once it is created.**

13 **Please share the Razorpay account details (API keys and webhook secret).**
Until then, payments run in test mode.

14 **Which email address and provider should send ADEM's emails?**
For example noreply@yourdomain.com, through your email provider.

15 **What GST details should appear on invoices (legal name, GSTIN, address)?**

16 **What is the website address (domain) for ADEM, and where will it be hosted?**

17 **Who will provide the Privacy Policy and Terms of Use text?**

18 **Do you confirm that existing test accounts and data can be removed (after a backup) for a fresh start?**

Features

19 **Do you approve these additions: GST invoices, email notifications, year-end class rollover, a support request form, and phone-friendly screens?**
Suggestion: approve all five.

20 **Should parents see the correct answers after the report, or only their score and marked sheet?**
Currently planned: score and marked sheet only; the answer key stays hidden.

Blockers: what is holding work back

These items must be in place before the related work can go ahead. Until then, work continues with test settings where possible.

Blocker	Affects	What happens until it is resolved
Backup and approval to remove existing test data	Phase 1	The new account system cannot be set up.
Email sending address and provider	Phase 1	No sign-up confirmation, invitations or password reset emails, so new users cannot join.
Google sign-in Client ID	Phase 1	Email and password sign-in works; Google sign-in waits.
Privacy Policy and Terms of Use text	Phase 1	Sign-up pages show placeholders; launch is not possible without them.
Credit prices and rules (questions 1 to 7)	Phase 2	Example values are used for testing.
Razorpay account details	Phase 2	Payments run in test mode only; no real money is collected.
GST details for invoices	Phase 2	Invoices are produced without GST details.
Pilot subjects, source books and a subject expert	AI track	The paper generator cannot be built or approved, so parent exams wait.
Website domain and hosting	Phase 6	The platform cannot go live; Google sign-in and email links need the final address.

Known issues and risks

Issue	Impact	How we are handling it
The scope of this release is large	HIGH	Work is split into a first release and a follow-up release. If time runs short, follow-up items move out, never the core flows.
Automatic papers are used without a person checking each one	HIGH	Papers are built from textbook content and checked automatically; an expert reviews samples before launch; parents can raise complaints about marks, with refunds.
Parents' phone photos may be poor	MEDIUM	A photo quality check before any credits are used, with guidance and a retake option.
Home exams cannot be fully supervised	MEDIUM	On-screen, timed papers that close automatically. We will not advertise exams as cheat-proof.

Issue	Impact	How we are handling it
Tamil textbooks are often hard for computers to read	MEDIUM	We use the same Tamil reading engine as for answer sheets, and check samples by hand.
Marking and paper creation share the same AI hardware	MEDIUM	Marking always takes priority; paper creation waits its turn. Speed is measured during testing.
Rules for children's personal data	MEDIUM	Parental consent at sign-up, a "delete my data" option, and automatic removal of uploaded files after exams.